EHR foundation models improve robustness in the presence of temporal distribution shift

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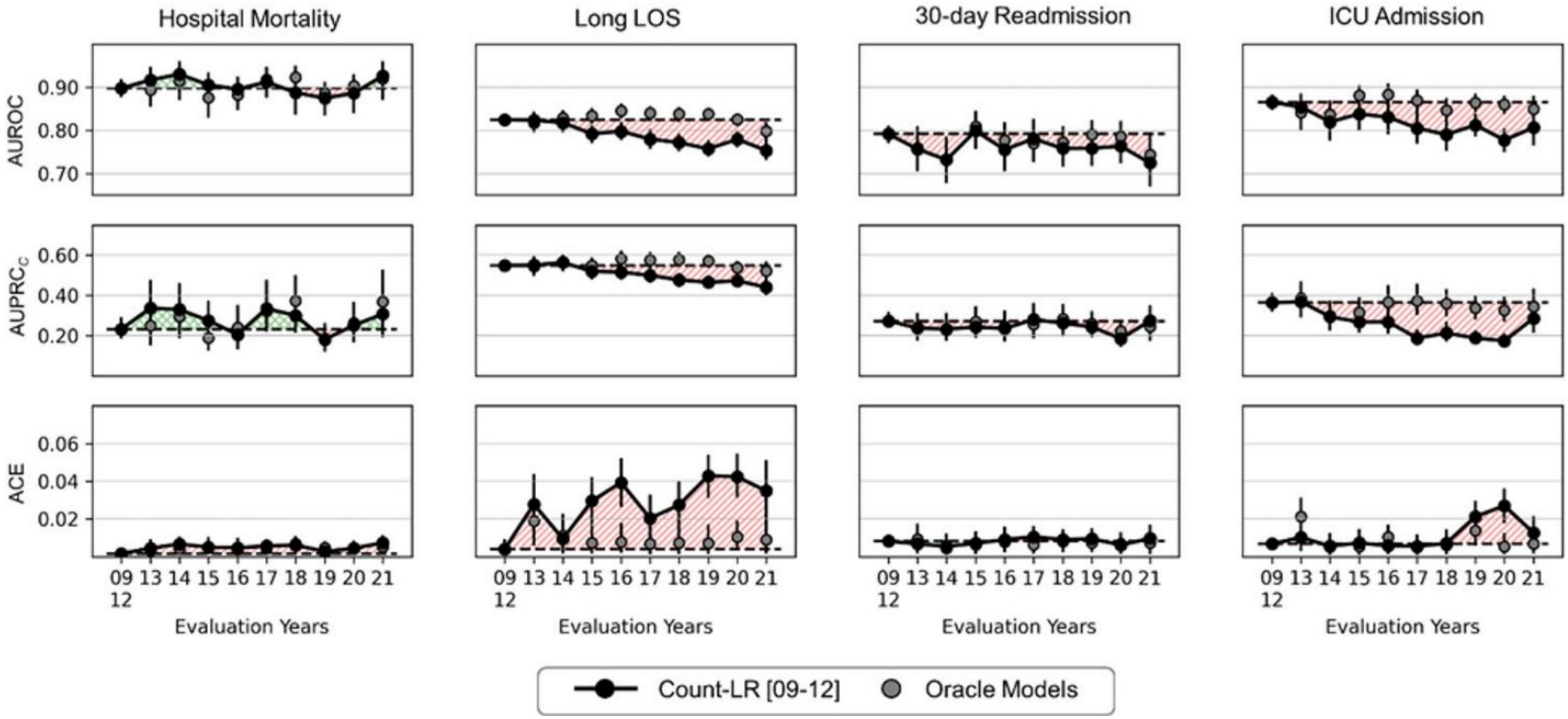
The Challenge: Clinical AI Models Age Poorly

- Clinical prediction models can lose performance when the data distribution changes over time.
- Healthcare data are non-stationary, so models trained on earlier years may degrade on later years. A model trained on 2015 data may not be accurate in 2023.

The impact of temporal distribution shift on count-based representations



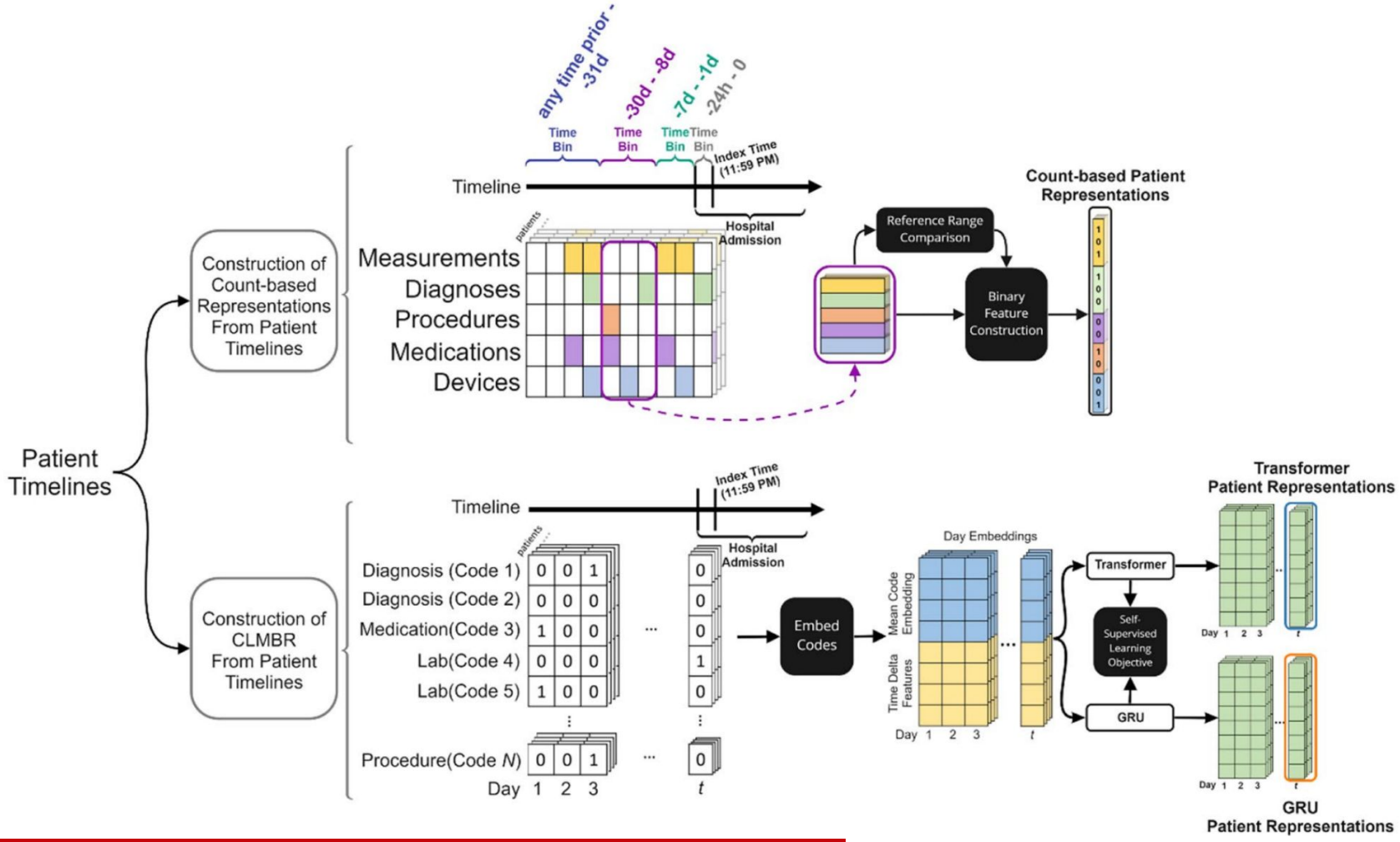
Dataset and Prediction Tasks

- Data come from the STARR repository at Stanford Medical Center, mapped to OMOP-CDM.
- The task cohort includes adult inpatient admissions, with one index admission randomly selected per patient.
- The four downstream tasks are **hospital mortality, long LOS, 30-day readmission, and ICU admission**.

Method

- The paper compares two patient representations: count-based features versus CLMBR representations.
- CLMBR treats a patient record as a time-ordered sequence of daily medical codes.
- After pretraining, the encoder is frozen and a logistic regression head is trained for each downstream task.

Method


Patient Timelines
Construction of Count-based Representations From Patient Timelines
Construction of CLMBR From Patient Timelines
any time prior - -31d
-30d - -8d
-7d - -1d
-24h - 0
Time Bin
Time Bin
Time Bin
Time Bin
Index Time (11:59 PM)
Timeline
Hospital Admission
Measurements
Diagnoses
Procedures
Medications
Devices
Reference Range Comparison
Binary Feature Construction
Count-based Patient Representations
Diagnosis (Code 1)
Diagnosis (Code 2)
Medication(Code 3)
Lab(Code 4)
Lab(Code 5)
Procedure(Code N)
Day 1 2 3
t
Embed Codes
Day Embeddings
Mean Code Embedding
Time Delta Features
Transformer
Self-Supervised Learning Objective
GRU
Transformer Patient Representations
GRU Patient Representations

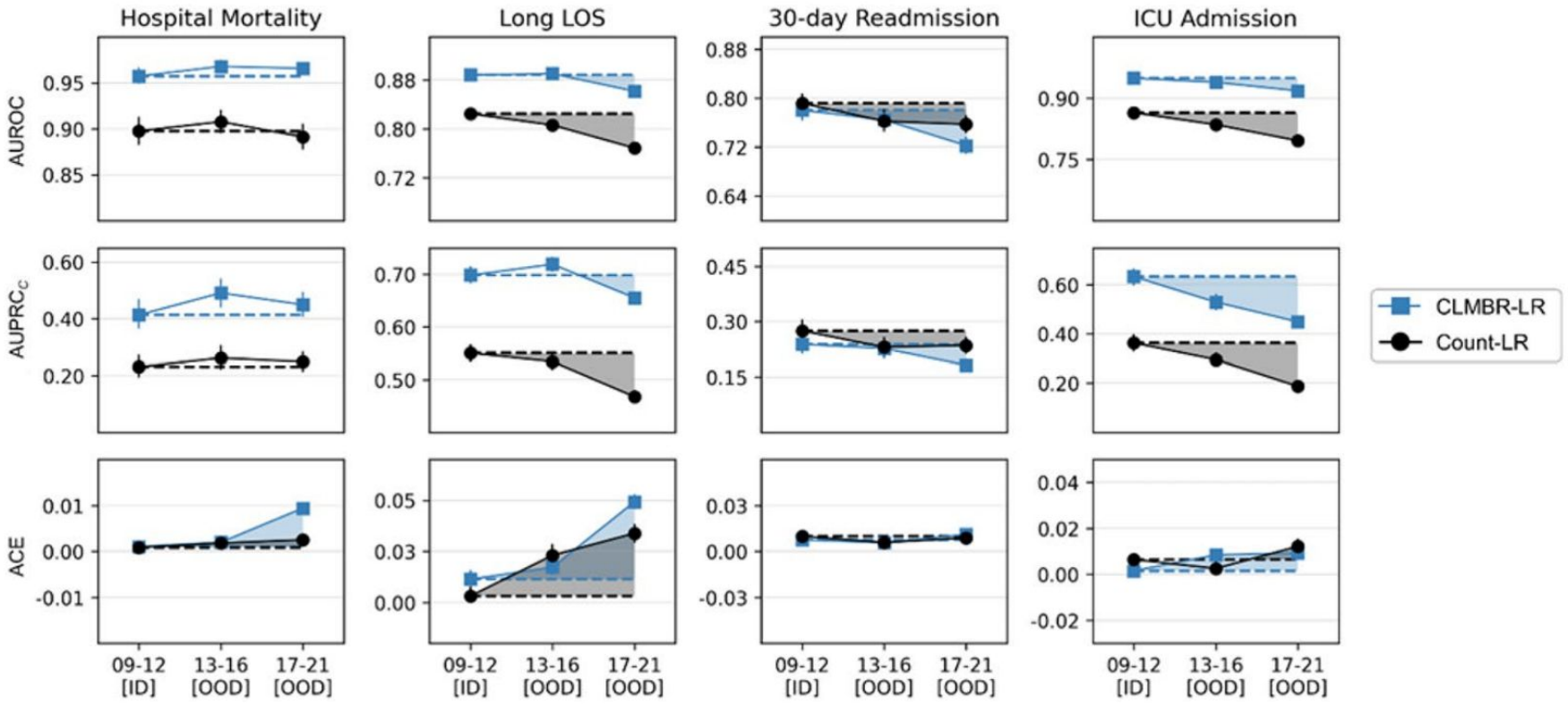
Method

- The pretrained encoder is either a Transformer or a GRU.
- Daily code sets are embedded, combined with temporal features, and mapped to an 800-dimensional patient representation.
- Pretraining uses an autoregressive next-day code prediction objective with binary cross-entropy loss; downstream adaptation uses L2-regularized logistic regression.

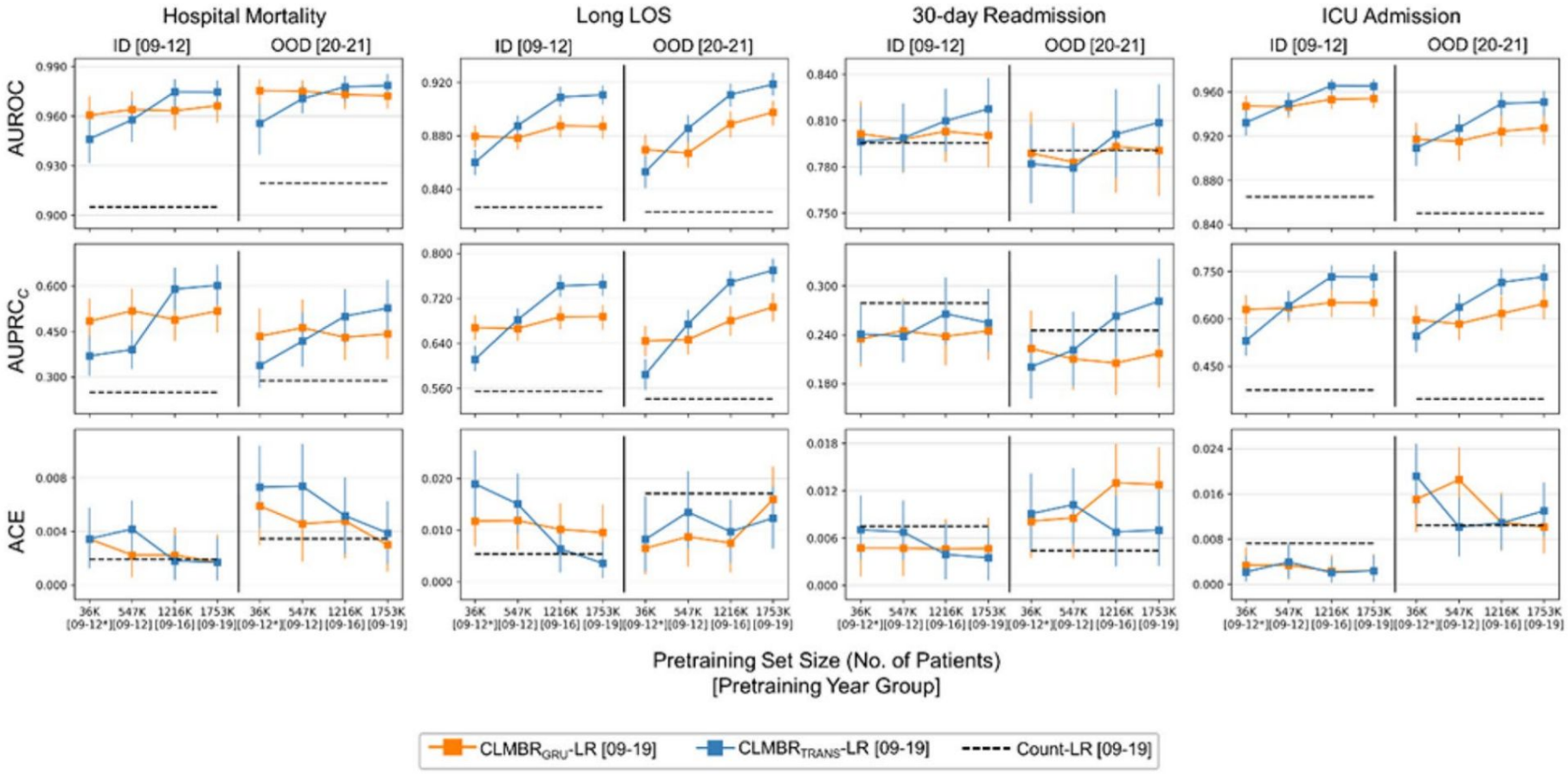
Experimental Design

- Experiment 1: compare CLMBR-LR with count-LR on ID years (2009–2012) and OOD years (2013–2016, 2017–2021).
- Experiment 2: study scaling behavior as pretraining size grows from 36K to 1.8M patients.
- Additional analyses examine pretraining contribution, end-to-end baselines, and clinical reclassification impact.

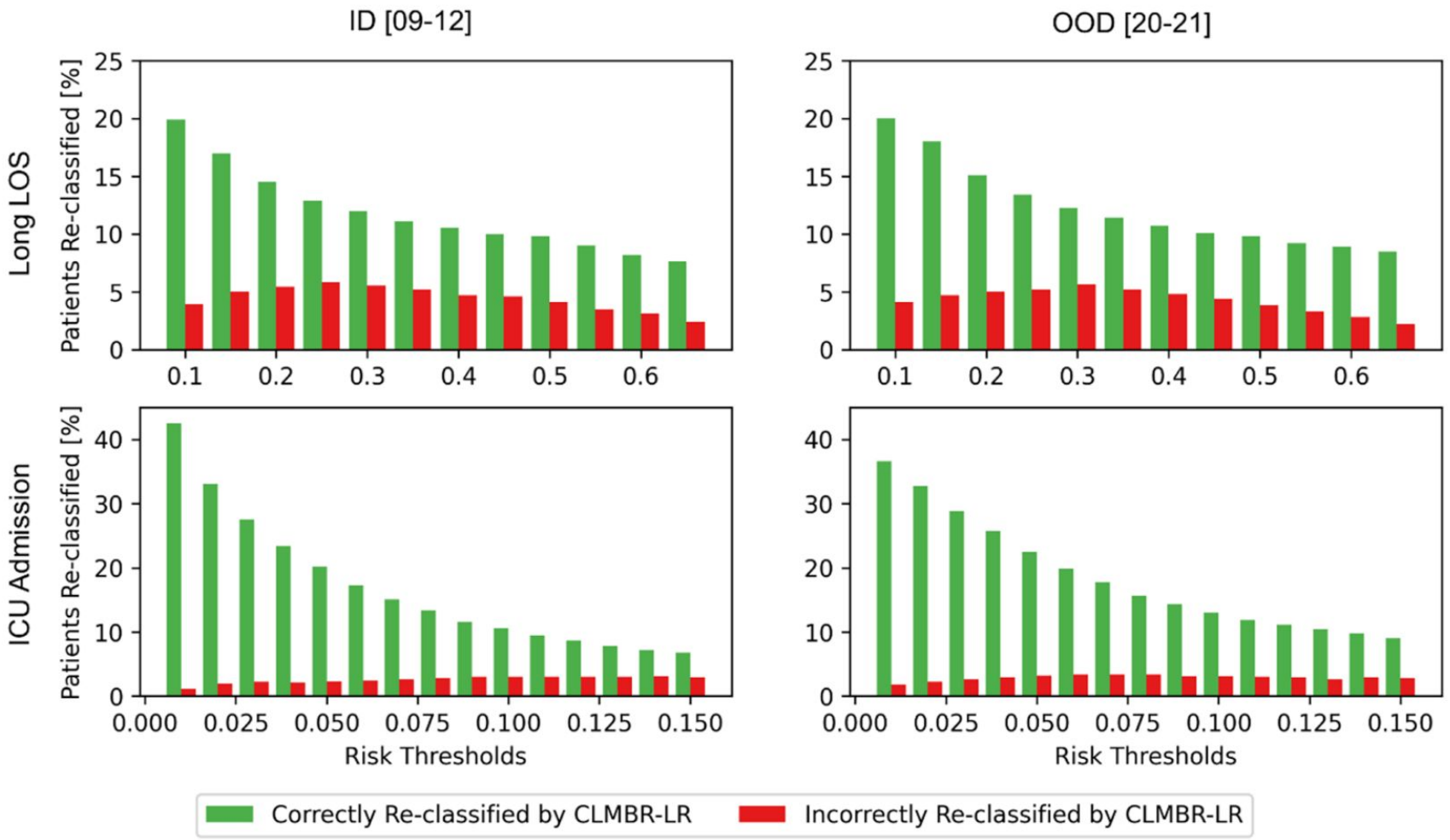
Main Result: CLMBR Often Degrades Less



GRU-based CLMBR works better at smaller pretraining sizes



Fewer Errors in Practice



Conclusion

Superior Performance

Models trained on CLMBR representations were better at discrimination and showed less performance decay over time compared to traditional count-based models.

Scaling is Key

The performance and robustness of transformer-based foundation models improved as the pretraining dataset size increased, confirming that scale is a critical factor.

A Robust Approach

Pretraining EHR foundation models at scale is a useful and promising strategy for developing clinical prediction models that perform well both now and in the future.

Limitations

Single-center EHR source, gains are task-dependent, and calibration can be worse or heterogeneous.